

Influence of shot peening on residual stresses and tribological behavior of cast and austempered ductile iron

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ABSTRACT

Austempered ductile irons exhibit an interesting combination of properties such as low cost, elevated strength, fatigue and wear resistance. In the present contribution, the influence of shot-peening on the dry-sliding wear behavior of as-cast and austempered ductile irons was evaluated. Specimens from a casting with an initial ferritic-pearlitic matrix were submitted to austempering at 320 °C for 60 min after which an ausferritic microstructure with 40 HRC and approximately 30% retained austenite was obtained. The wear and friction behaviors were assessed by applying ball-on-disc tests using a hard metal (WC) counter-body on as-cast and austempered materials before and after shot-peening and after removal of the rough surface layer created by the shot-peening process. The microstructure of the tested materials was evaluated by X-ray diffraction, optical and scanning electron microscopy. After shot-peening, the austempered samples experienced an elevated increase in surface hardness because of the formation of strain-induced martensite, but a reduction in wear resistance caused by the increase in surface roughness. By removing 20 μm from the surface of the shot-peened samples, it was possible to reduce roughness while preserving the hardened layer, which was found to improve wear resistance.

1. Introduction

Austempered Ductile Irons (ADI) are metal alloys obtained by austempering conventional Ductile Irons (DI) and offer an interesting combination of properties which include elevated mechanical strength, ductility, fatigue resistance, wear resistance, as well as low cost in comparison with steels of equivalent hardness. These properties are due to the characteristic ADI microstructure, which consists of graphite nodules dispersed in a matrix of acicular ferrite and retained austenite ("ausferrite"). Due to its excellent mechanical properties, ADI's are being employed or considered as structural materials in civil construction, automotive components, agricultural and mining equipment, among other applications [1–3].

The austempering of ductile cast irons is illustrated schematically in Fig. 1, which presents the processing route superimposed on the material's isothermal transformation diagram. First, heating until full austenitization is achieved, following rapid cooling down to the austempering temperature (T_A), which lies in the bainite formation range (usually between 290 and 420 °C). From this point, austenite is gradually transformed into ferrite. The carbon content in the remaining

austenite is increased and if the material is quenched from Stage I to room temperature, the desired ausferrite microstructure will be formed, in which retained austenite is stabilized at room temperature due to the elevated amount of carbon in solution. If the material is kept at T_A long enough, complete transformation to bainite will eventually be achieved, with lower ductility and toughness [4]. The target mechanical properties of ADI occur between the end of Stage I and the start of Stage II of the reaction presented in Fig. 1, which gives the process a characteristic "processing window" [5–7]. Normally, for higher T_A values, ductility and toughness are increased at the expense of hardness [4,8,9].

An important process which is largely applied for increasing fatigue resistance of mechanical components is shot-peening, in which high velocity shots with elevated hardness are bombarded against the workpiece, causing plastic deformation in the surface and sub-surface regions. This leads to strain-hardening and compressive residual stress fields, which hinder crack nucleation and growth, increasing fatigue resistance [10]. Because of its potential for increasing fatigue resistance, the application of shot peening to ADI has been considered regarding surface residual stresses and fatigue strength [11–15], but relatively less attention has been directed to investigate tribological aspects of the

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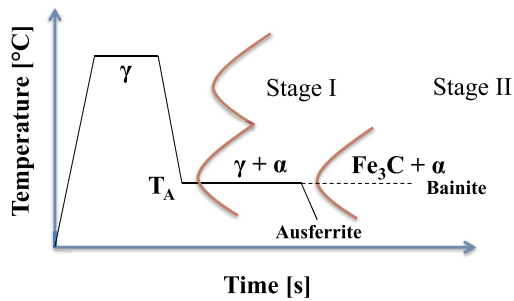


Fig. 1. Schematic representation of the austempering process.

Table 1

Chemical composition of the materials used in the present study (wt.%).

C	Si	Mn	P	S	Cu	Cr	Mo	Fe
2.60	2.50	0.34	0.07	0.04	0.04	0.04	0.04	Bal.

shot-peened surface in these materials [14,16,17]. In the case of ADI, application of shot-peening may also lead to phase transformations because of the elevated amount of retained austenite [18,19].

The strain-induced transformation of austenite to martensite further increases surface hardness, as has also been reported for case-hardened alloy steel [20] as well as austenitic stainless steels submitted to ultrasonic and laser-shock peening [21], severe shot-peening [22] and machining processes [23,24]. As such, the tribological evaluation of shot-peened ADI is interesting because of the potentially negative influence of increased surface roughness on wear resistance in comparison to the expected positive effects of increased surface hardness and compressive stresses. Based on these premises, in the present work an investigation of the influence of shot-peening on the wear behavior of ADI was performed. Samples of unalloyed ductile irons (with mostly ferritic microstructure) in the as-cast and austempered state were submitted to shot-peening and wear testing was conducted on both peened and non-peened specimens.

2. Experimental procedure

2.1. Materials

The samples used in all experimental analyses were cut from standard Y-blocks (following ASTM A536) by conventional milling and their chemical composition is presented in Table 1. The cast materials were selectively submitted to austempering heat treatments and shot-peening in order to obtain specimens in the following conditions: as-cast (AC), as-cast and shot-peened (AC + SP), austempered (ADI) and austempered and shot-peened (ADI + SP). Prior to any processing by either austempering or shot-peening, all materials were ground with SiC paper to normalize surface roughness conditions (see Fig. 8). The microstructure of the as-cast materials was formed of a ferritic-pearlitic matrix with average graphite nodularity between 85 and 90%, class I graphite distribution and class 7 nodule size.

2.2. Austempering and shot-peening processes

The austempering heat treatment was performed by austenitization at $900 \pm 3^\circ\text{C}$ for 60 min for complete microstructure homogenization in a salt bath to avoid surface decarburization followed by rapid cooling in salt bath at $320 \pm 5^\circ\text{C}$. The austempering temperature of 320°C was selected based on a previous investigation on the processing window of a similar ductile iron composition [25]. In order to evaluate the evolution of austenite content during heat treatment, samples were austempered for 5, 10, 15 and 60 min before being quenched to room temperature in a

brine solution. After initial analysis, only samples submitted to 60 min austempering at 320°C were chosen for further examination. Austempered (ADI) and as-cast (AC) samples were then submitted to shot-peening treatments. The shot-peening intensity was determined by the Almen method using $75 \times 19 \times 1$ mm sheets of AISI 1070 steel with hardness between 44 and 50 HRC. The procedures were carried out in Pangborn PE1107 shot-peening equipment, with 18A Almen intensity, 30 s exposure per sample and 100% coverage. Spherical S230 grade shots of 0.6 mm diameter were used, with 60 ± 2 HRC hardness. The surface roughness of the AC, AC + SP, ADI and ADI + SP samples was evaluated by using a TIME TR210 digital roughness tester. A $0.8 \mu\text{m}$ cutoff was used for the unpeened specimens while a $2.5 \mu\text{m}$ cutoff was applied for the peened specimens and 5 measurements were performed for each surface condition.

2.3. Characterization and residual stress analysis

Microstructure characterization was performed by Optical Microscopy (OM) and Scanning Electron Microscopy (SEM). Sample preparation was performed by following standard metallographic procedures, which involved grinding in SiC paper and polishing down to a $1 \mu\text{m}$ finish in diamond suspension. Microstructure features were revealed by etching in 2% Nital solution.

The austempering process was analyzed by X-ray Diffraction (XRD) in order to follow the phase transformations that take place during heat treatment and also to assess the residual stresses before and after shot-peening. Residual stress analysis was carried out on austempered samples (ADI and ADI + SP) by applying the conventional $\sin^2\psi$ method using $\text{Co K}\alpha$ radiation (0.178897 nm wavelength). The ferrite-211 (elastic constants $S_1 = -1.268 \text{ MPa}^{-1}$ and $\frac{1}{2}S_2 = 5.804 \text{ MPa}^{-1}$) and austenite-311 (elastic constants $S_1 = -1.545 \text{ MPa}^{-1}$ and $\frac{1}{2}S_2 = 6.439 \text{ MPa}^{-1}$) reflections were analyzed in each case, and measurements were taken at 0 and 180° azimuthal angles.

2.4. Hardness evaluation

Depth resolved Vickers microhardness measurements were performed on cross sections of all samples (AC, AC + SP, ADI and ADI + SP) used in the present work. The measurements were performed using a Shimadzu HMV-2T microhardness tester with a 10 gf load and 20 s holding time. For each sample, 3 line profiles were determined, and all measurements were performed on the metallic matrix (care was taken to avoid indentations in graphite nodules). Microhardness measurements along the cross-section of the samples allowed the assessment of the in-depth hardness gradient expected to be induced by the shot-peening process. The step size used for obtaining the line profile was of $25 \mu\text{m}$.

2.5. Tribological behavior

Dry sliding ball-on-disc tests were performed on a conventional Microtest M/NI-1E tribometer for determination of friction coefficients and mass loss. A limitation of this work is that the tests were conducted with a fixed set of parameters without a specific industrial application in mind. This was done in order to favor the examination of different surface microstructures. An applied load of 15 N was employed, with a 17 mm radius and 0.2932 m/s sliding speed which resulted in a total sliding distance of 3600 m. The counter ball used was a hard metal (WC) sphere, with 6.35 mm diameter and initial mass of 2.009 g. The tests were repeated once for each experimental condition. Wear volume was determined by separately weighing the mass of both disc (sample) and the hard metal sphere before and after testing of each specimen (for each experiment, a new disc and a new sphere were used). The sphere material was chosen because its hardness is significantly higher compared to the DI and ADI samples (before and after shot-peening) and should thus exhibit similar behavior regardless of test material. The ADI + SP samples were tested both in the shot-peened condition and after removal

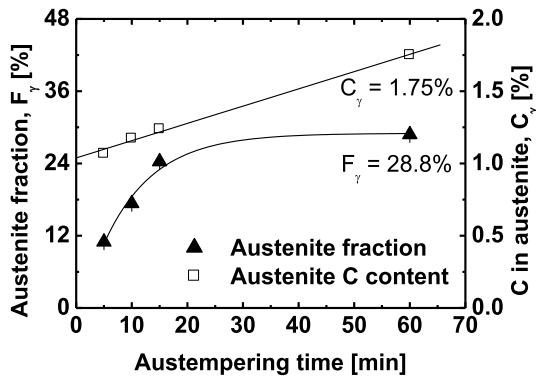


Fig. 2. Evolution of austenite fraction (F_γ) and austenite carbon content (C_γ) with austempering time at 320 °C.

of 20 μm by surface milling (ADI + SP + M). This additional analysis was performed in an attempt to separately evaluate two surface effects of shot-peening on the wear behavior: hardening (with an expected increase in wear resistance) and roughening (which is expected to decrease wear resistance). After the ball-on-disc tests, the worn surfaces of the ADI + SP and ADI + SP + M samples were examined by optical profilometry.

3. Results and discussion

3.1. Microstructure and residual stress evaluation

The evolution of retained austenite during austempering between 5 and 60 min was followed by XRD and the results are presented in Fig. 2, which shows plots of the austenite fraction (F_γ) and the austenite carbon content (C_γ). The amount of carbon dissolved in austenite was determined from the variation in lattice parameter “ a ” of the FCC structure of Fe by using Eq. (1):

$$C_\gamma = (a - 3.548) / 0.044 \quad (1)$$

For short austempering times at 320 °C (5 and 10 min), limited ferrite formation takes place and there is not sufficient carbon enrichment to stabilize austenite upon cooling to room temperature. As the treatments proceeds (from 15 min onwards), austenite carbon content increases to a level which allows formation of a higher fraction of stabilized austenite in the final microstructure [26]. It also appears that the amount of retained austenite is stabilized at close to 30% (the value of 28.8% was obtained after 60 min austempering at 320 °C). For this reason, the austempering time of 60 min was selected for further analysis and shot-peening.

The microstructure of the samples austempered for 60 min is examined in Fig. 3(a and b) by SEM/EBSD. Phase maps are presented in Fig. 3(a) while inverse pole figure maps are shown in Fig. 3(b) in order to highlight the individual grains in the microstructure. The quantitative phase analysis performed by EBSD revealed fractions of ferrite, austenite and graphite of 0.207, 0.639 and 0.154, respectively. In Fig. 3(c) the corresponding confidence index map is presented, in which it is possible

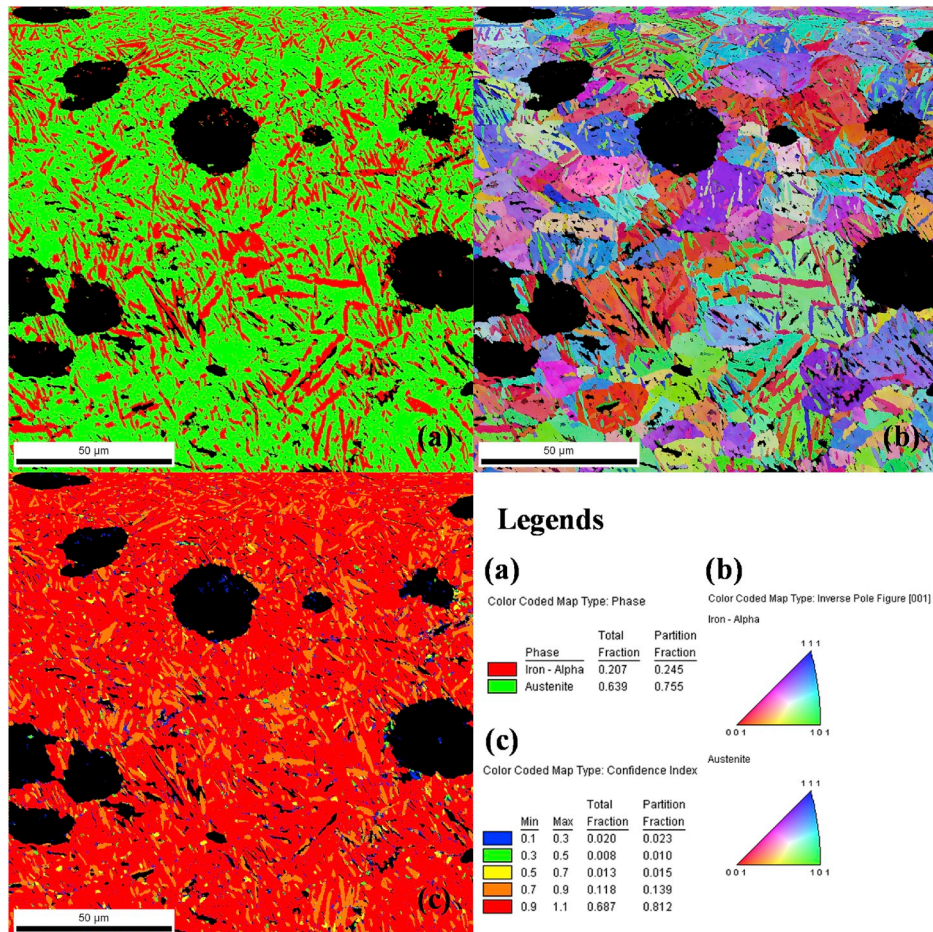


Fig. 3. SEM/EBSD analysis performed after austempering at 320 °C for 60 min: (a) phase map, (b).

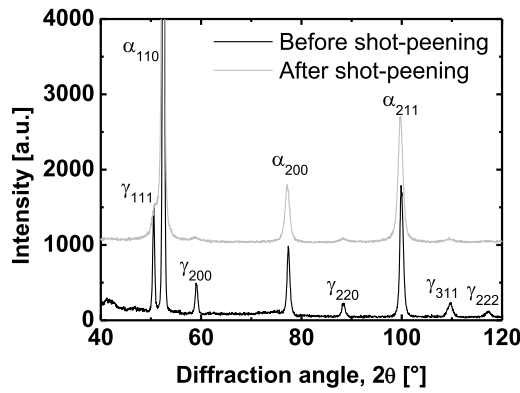


Fig. 4. Diffraction patterns obtained from ADI samples before and after shot-peening.

to notice that the identification of the metallic phases could be reliably performed, but no patterns could be analyzed for the graphite nodules. As such, the values obtained for the graphite fraction were determined by exclusion. It was possible to confirm the presence of the typical ausferrite microstructure, composed of relatively high levels of retained austenite with acicular ferrite. The fraction of each constituent (ferrite/austenite) determined by EBSD is not compatible with the values obtained XRD, but this can be attributed to the fact that information obtained from EBSD are local while the volume probed by XRD is significantly larger and more representative of the bulk of the material.

In Fig. 4, XRD data obtained from samples austempered for 60 min before and after the application of the shot-peening treatment are compared. It is possible to notice that the shot-peening process leads to changes in the amount of retained austenite, which decreases from 28.8% to 5.0% because of martensite formation. The results of the residual stress analysis are presented in Fig. 5(a)–(b) in the form of $\sin^2\psi$ plots for the ferrite and austenite phases, respectively, and the residual stress values obtained are registered in Table 2. During the XRD residual stress evaluation, it was possible to notice that the integral breadth of the analyzed peaks increased with shot-peening from approximately 0.86 and 1.16 to 1.44 and 1.40 for ferrite and austenite, respectively. The observed increase in XRD line broadening is consistent with two effects of the shot-peening process, namely lattice strain caused by plastic deformation and reduction in crystallite size caused by grain refinement [22,27].

The residual stress values presented in Table 2 reveal that differences obtained from measurements at 0 and 180° azimuthal directions remain between the limits of the standard error, which means that negligible shear components are expected in the surface residual stress state. After austempering, low magnitude tensile stresses are present at the surface,

which become compressive after shot-peening. It is worth noticing that the magnitude of the residual stresses in austenite, after shot-peening, is significantly larger than in ferrite. This can be explained by considering that with shot-peening most of the initial austenite is transformed to martensite (Fig. 4), with a volume expansion that is accommodated by elastic strains in the surrounding microstructure (in the present case, the small amount of residual austenite still present after surface bombardment) [11]. According to Ebenau et al. [11], the mean stress increment ($\Delta\sigma$) can be estimated as a function of the reduction in retained austenite (ΔRA), the materials elastic modulus (E) and the local elastic strain (ε) corresponding to a third of the isotropic volume increase:

$$\Delta\sigma = \Delta RA \times E \times \varepsilon \quad (2)$$

By considering ΔRA , and E equal to 23.8% (obtained from XRD) and 210 GPa, respectively, and isotropic volume increases between 2.1% [11] and 4.3% [28] the mean stress increment could be estimated in the range of 350–720 MPa, consistent with the differences in residual stresses observed in Table 2 between ferrite and austenite.

The microstructure of the AC, AC + SP, ADI and ADI + SP samples can be observed at low resolution near the surface region in Fig. 6(a–d), respectively. From Fig. 6(a) it is possible to notice that the AC microstructure is composed mostly of ferrite with pearlite islands. After shot-peening, Fig. 6(b), the initially equiaxed ferrite grains become visibly deformed near the surface. The ADI microstructure is difficult to resolve by analyzing low resolution micrographs but it is nevertheless possible to discern relatively large volumes of retained austenite near the surface region in Fig. 6(c), below which the microstructure is predominantly ausferrite, as shown in higher resolution in Fig. 3. By comparing Fig. 6 (d) and (b) it is possible to notice that the shot-peening process had a larger impact on the as-cast material, by the relative size of the surface indentations produced during the process. This behavior can be explained based on the larger hardness difference between shot material (approximately 60 HRC or 780 HV) and alloy (approximately 250 HV in the as-cast condition and 400 HV after austempering).

3.2. Hardness and surface roughness evaluation

The results of the microhardness measurements performed on the as-

Table 2

Residual stress values obtained before and after shot-peening (the error values reflect the linear fit error from Fig. 5).

Phase	Azimuthal angle	ADI	ADI + SP
α -Fe	0°	120 ± 10 MPa	−440 ± 30 MPa
α -Fe	180°	110 ± 10 MPa	−390 ± 20 MPa
γ -Fe	0°	60 ± 20 MPa	−870 ± 150 MPa
γ -Fe	180°	60 ± 20 MPa	−1100 ± 60 MPa

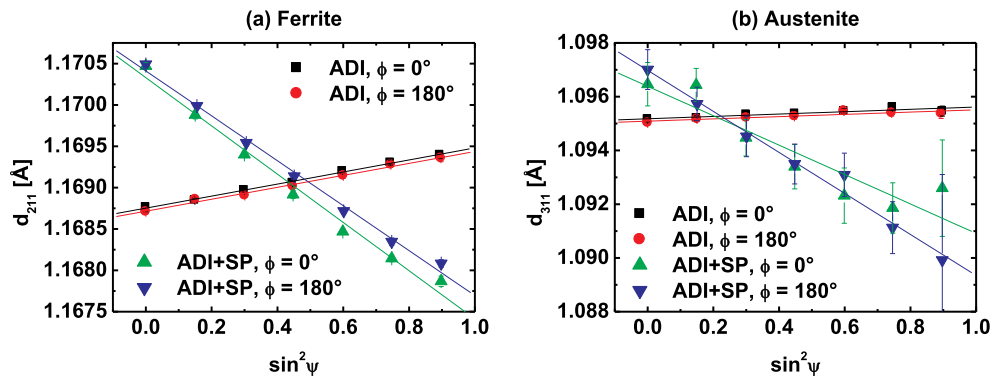


Fig. 5. Residual stress analysis performed on ADI and ADI + SP samples for the (a) ferrite and (b) austenite phases (the error bars represent the error associated with the determination the center of each diffraction line).

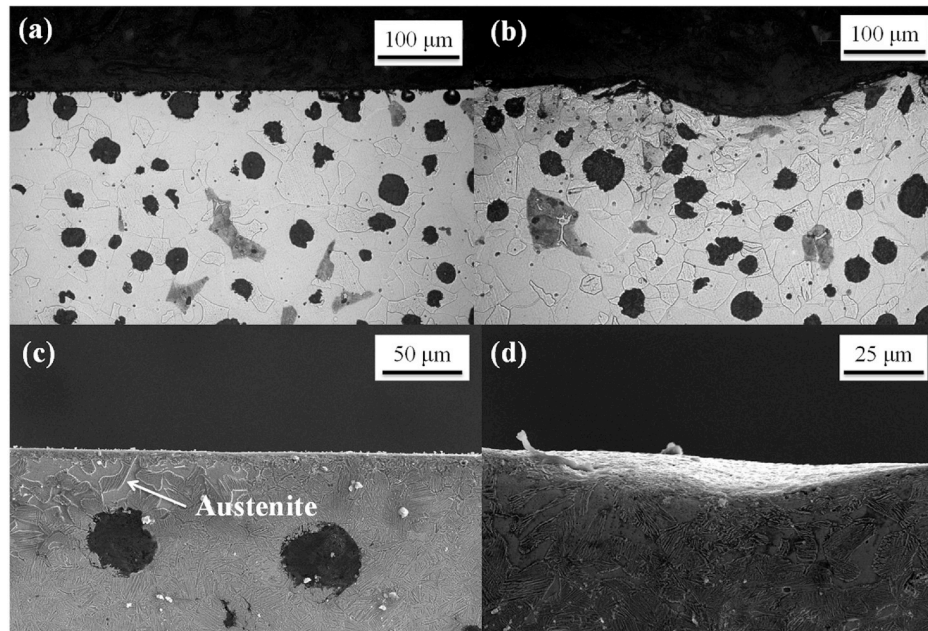


Fig. 6. Microstructure of the near surface regions the AC sample (a), AC + SP (b), ADI (c) and ADI + SP (d).

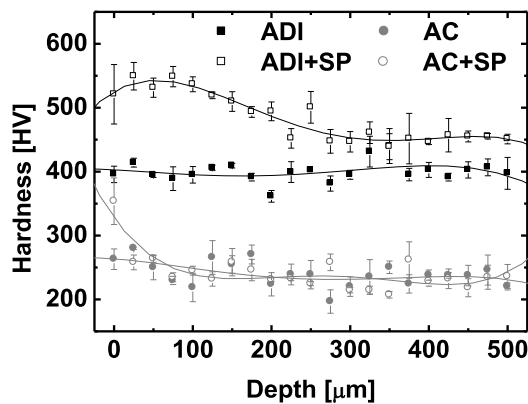


Fig. 7. Microhardness depth profile obtained before and after shot-peening on as-cast and austempered ductile iron (the error bars reflect the standard deviation of 3 measurements).

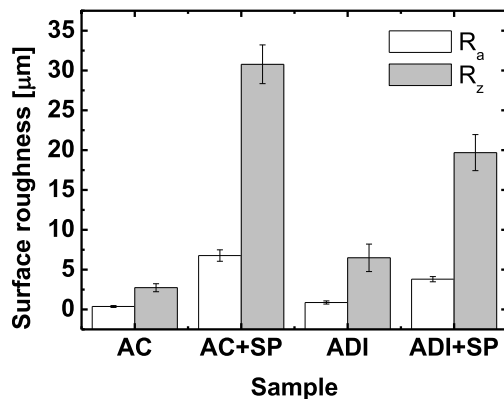


Fig. 8. Surface roughness parameters obtained before and after shot-peening on as-cast and austempered ductile iron (the error bars represent the standard deviation of 5 measurements).

cast specimens (AC and AC + SP) and the austempered specimens (ADI and ADI + SP) are presented in Fig. 7. Concerning the as-cast samples, the results presented in Fig. 7 reveal an increase in surface values from close to 263 ± 16 HV to nearly 354 ± 36 HV and a limited strain-hardened zone of approximately between 25 and 50 μm in depth after which hardness values stabilize at ~ 230 HV for both materials. No particular trend could be identified in the non-peened material. In turn, the heat-treated samples exhibited significantly higher hardness values due to the austempering process in comparison to the as-cast samples. The highest hardness value registered for the ADI + SP sample was 549 ± 21 HV at a 25 μm depth from the surface while for the ADI sample a maximum hardness of 440 ± 27 HV 350 μm below the surface. Considering the XRD results presented in Fig. 4 it is possible to conclude that the increase in hardness is caused by the strain induced transformation of austenite to martensite. The values obtained here are comparable to results found in previous investigations [16,29].

The results obtained from the surface roughness (R_a and R_z) measurements are presented in Fig. 8. The shot peened samples exhibited highest roughness levels, as expected, due to plastic deformation the caused by hard particle impacts on the surface. The cast material (AC) exhibited average $0.37 \mu\text{m}$ R_a which evolved to $6.77 \mu\text{m}$ R_a upon shot peening (AC + SP), whereas the austempered material (ADI) exhibited originally $0.87 \mu\text{m}$ roughness which increased to $4.23 \mu\text{m}$ with shot-peening. The larger increase in roughness on the AC specimen is due to the softer surface relative to the ADI sample, as noted in Fig. 6(b,d) which revealed more pronounced indentations on the as-cast materials relative to the austempered condition. Hence, higher levels of plastic deformation were experienced in the as-cast material with shot-peening, increasing both quantity and amplitude of surface peaks. Further corroboration for this interpretation can be found when noticing that for the non-peened samples, the austempering process led to an increase in surface roughness – which can be attributed to the presence of a surface oxide scale created during the heat treatment process. After application of the shot-peening process, however, the situation is inverted, with the AC material presenting the roughest surface finish.

3.3. Tribological behavior

The sliding wear resistance of the AC, AC + SP, ADI and ADI + SP specimens was evaluated by performing ball-on-disc tests. The ADI + SP

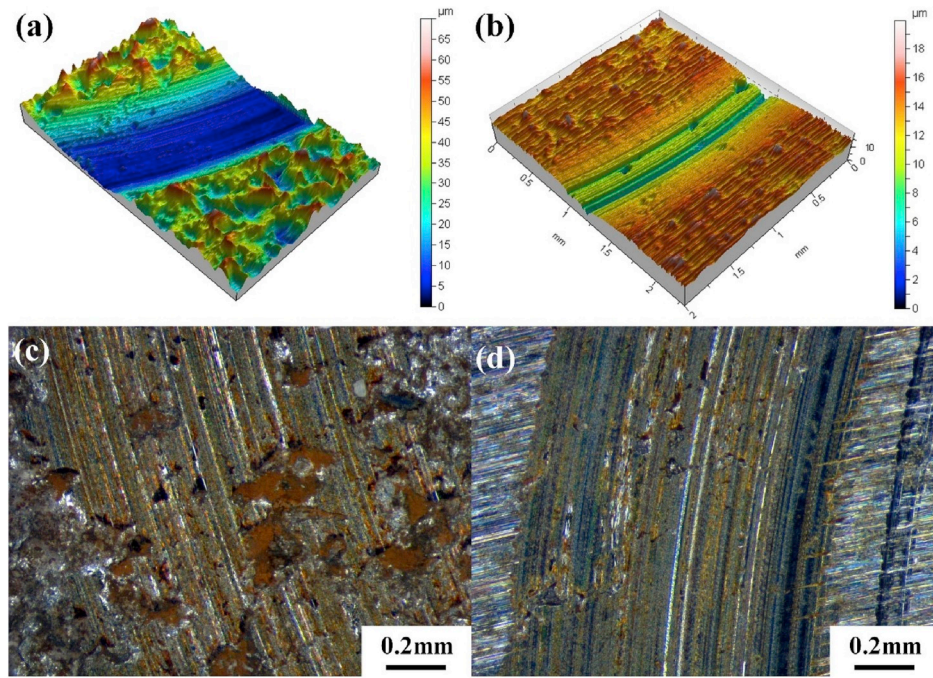


Fig. 9. Examination of the wear tracks obtained after ball-on-disc tests on the ADI + SP (a,c) and ADI + SP + M (b,d) by optical profilometry (a,b) and optical microscopy (c,d).

samples were evaluated in the shot-peened condition and after removal of 20 μm by surface milling (ADI + SP + M) with the objective of eliminating the rough surface layer (Fig. 8) while keeping the hardened layer obtained after shot-peening (Fig. 7). The mass loss values obtained for the AC, AC + SP, ADI, ADI + SP and ADI + SP + M specimens were, respectively, 0.8 ± 0.2 , 25.2 ± 0.4 , 8.4 ± 2.7 , 12.9 ± 6.2 and 3.1 ± 0.1 mg. Because of the large difference in hardness in comparison to the ADI discs, the hard metal spheres used in the tests exhibited negligible wear.

The wear behavior of the as-cast samples was more heavily influenced by the shot-peening process in comparison to the austempered materials, in agreement with the observations made concerning the surface roughness results (Fig. 8). Concerning the ADI samples, the shot-peening process was found to produce an increase in mass loss in agreement with [13], probably in connection with the increase in surface roughness and hardness caused by the strain-induced transformation of austenite to martensite. The harder (less ductile) and more pronounced surface asperities are more easily fractured and removed

during wear testing. Hence, by removing the larger asperities (ADI + SP + M), it was possible to significantly reduce mass loss and wear resistance was found to be improved in comparison to both ADI and ADI + SP samples, in agreement with Palacios et al., who performed a grinding procedure following the application of shot-peening on 6063 Al-alloy [30]. The results obtained here show that the beneficial effects of shot-peening in terms of compressive residual stresses and hardness were counterbalanced by the increase in surface roughness, in agreement with previous reports [16,29] and that this effect could be mitigated by controlled removal of the rough surface layer.

The surface of the wear tracks on the ADI + SP and ADI + SP + M samples is analyzed in Fig. 9 by optical profilometry and optical microscopy. The observations confirm the trends indicated by the mass loss measurements, as the width of the track formed on the ADI + SP sample is visibly larger compared to the ADI + SP + M material, indicating a larger volume loss. Indeed, depth profile scans performed across the wear tracks indicated average areas 32600 ± 4500 and $1980 \pm 300 \mu\text{m}^2$

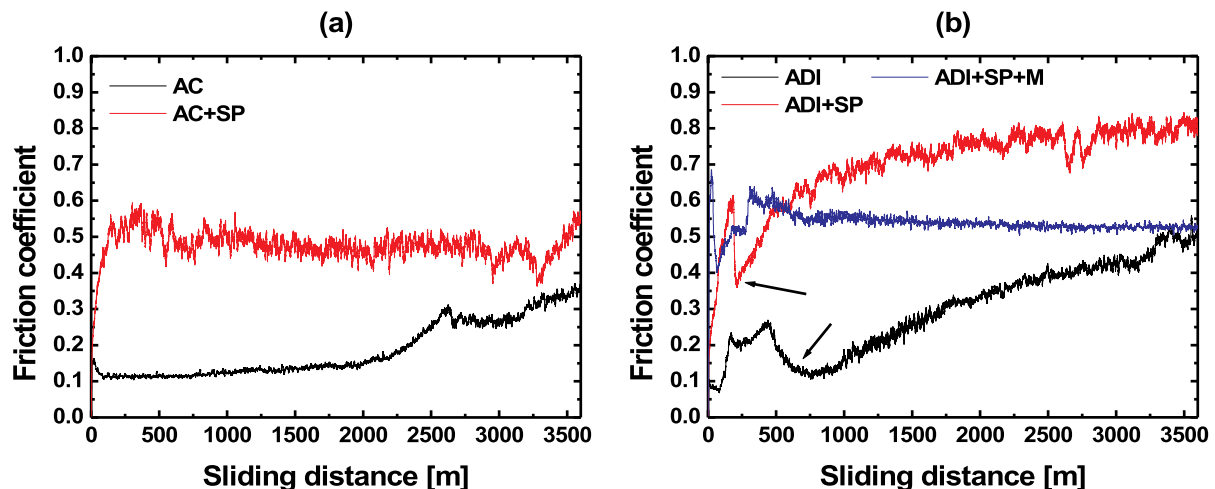


Fig. 10. Evolution of friction coefficient as a function of sliding distance for the (a) as-cast and (b) austempered materials.

for the ADI + SP and ADI + SP + M samples, respectively. The tracks exhibit ploughing marks, consistent with previous examinations of wear in ADI [31], red-brown spots, signs of oxidation taking place during the dry-sliding tests due to frictional heat generation [32] and pits [16,33]. These features can be identified on both surfaces, indicating that similar wear mechanisms take place. The difference, however, resides on the intensity of the surface damage, which is clearly larger on the ADI + SP sample.

The evolution of friction coefficient as a function of sliding distance for the as-cast (AC and AC + SP) and austempered (ADI, ADI + SP, ADI + SP + M) is presented in Fig. 10(a) and (b), respectively. It can be observed that the shot-peened samples (AC + SP and ADI + SP) presented superior friction coefficient relative to the unpeened materials, revealing a positive correlation with the mass loss values obtained. In addition, in the initial stages of the test, the shot-peened samples presented a similar evolution concerning friction coefficient – a sharp rise to values between 0.55 and 0.6. For the ADI + SP specimen the value is then reduced, as indicated by the arrow in Fig. 10. Since a similar drop was also noticed for the unpeened ADI material, which would be consistent with the removal of the oxide scale produced during the heat treatment process, present on the material surface before the dry-sliding tests. The initial friction behavior observed for the AC + SP and ADI + SP samples appears thus to be influenced by the higher surface roughness created by the shot-peening process. After the initial stages the friction coefficient for the as-cast materials remains stable but for the ADI samples gradually increases throughout the remainder of the test with similar rates.

It is worth noticing that removal of the layer by milling (ADI + SP + M) led to a reduction of the friction coefficient between sample and WC counter-body in comparison to the shot-peened material (ADI + SP) after approximately 500 m sliding distance. The behavior illustrated in Fig. 10(b) is consistent with the superior surface finish provided by the milling process and with the reduced overall mass/volume loss observed for the ADI + SP + M sample.

4. Conclusions

The purpose of this research was to investigate the influence microstructure and surface modifications induced by shot peening on the wear behavior of austempered ductile iron. The approach used to examine this problem involved microstructure characterization and tribological evaluation by means of ball-on-disc tests using a WC counter-body. As a result of this work, the major findings concerning the effects of shot-peening on wear of austempered ductile iron were:

- Shot peening led to transformation of retained austenite to martensite, with a hardness increase;
- The increase in hardness was insufficient to overcome the detrimental effect of increased surface roughness induced by shot peening on the wear behavior;
- Removal of the rough surface layer controlled milling ($\sim 20\ \mu\text{m}$) was found to preserve the positive effects on wear resistance produced by the increase in hardness.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wear.2019.203099>.

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